

# JuiceFlow Factory Manager & Customer Store

A comprehensive, top-to-end technical manual and operational guide for the complete JuiceFlow ecosystem: featuring industrial food manufacturing ERP, live Supabase PostgreSQL cloud backend synchronization, IoT plant monitoring, AI demand analytics, native background push notifications, and a direct-to-consumer automated juice storefront.

📱 **Platform:** Flutter (Android / iOS / Web) ☁️ **Backend:** Supabase PostgreSQL & SQLite

📖 **Enterprise:** FSSAI & HACCP Compliant 🌐 **Language:** English & தமிழ் Bilingual

🔔 **Background:** Native Android Notifications

## 1. System Architecture & Dual-Portal Paradigm

JuiceFlow is built with a unified dual-portal architecture that bridges the entire beverage supply chain into a single cohesive mobile application:

### 1 🏭 **Factory Producer & Operations Portal**

Designed for the factory owner, production supervisors, lab chemists, and inventory controllers. Manages raw fruit procurement, Brix sugar titrations, HTST pasteurization telemetry, batch traceability certificates, worker rosters, and cold-chain reefer van dispatch.

### 2 🛒 **Customer Storefront & Live Auto-Billing Portal**

Designed for retail consumers, cafes, supermarkets, and catering clients. Enables selecting fresh juice varieties, specifying custom bottle quantities or bulk crates, instant auto-generated tax billing with 5% GST, multi-mode digital payments (UPI, Cards, COD), and live order status tracking.

### 3 ☁️ **Live Cloud Backend & Offline-First SQLite Synchronization**

Bi-directional data synchronization with Supabase PostgreSQL cloud. When online, all production batches, orders, inventory, and staff logins sync instantly across all devices. When offline, local SQLite engine continues operations seamlessly without downtime.

## 2. Interactive 3D Tree-to-Tumbler Juicing Experience

When the application launches, users are greeted with a realistic physics-driven simulation of fresh fruits falling from an orchard tree canopy directly into a juicing vortex, filling a crystal tumbler with dynamic liquid waves and foam.

Customers

Search customer name, shop, phone...

Grand Residency Hospitality

Hotel & Resort • Contact: Hotel Grand Residency

+91 98840 22334 • Bypass Road, Dindigul - 624004

₹0.00

All Clear

Southern Distributor Network

Distributor • Contact: Southern Beverages Wholesale

+91 99420 55112 • Wholesale Market Complex, Trichy - 620008

₹34,000.00

Pending Due

Sundar Retail Mart Pvt Ltd

Supermarket • Contact: Sundar Stores (Supermarket)

+91 94432 11098 • 14 Cross Road, Madurai - 625001

₹12,500.00

Pending Due

Vasantham Retail Group

Retailer • Contact: Vasantham Department Stores

+91 97500 88210 • Main Bazaar, Thani - 625531

₹6,800.00

Pending Due

+ Add Customer

SPLASH & VISUAL ENGINE

Interactive 3D Fluid Pour Simulation

Simulates dynamic liquid mechanics with real-time fruit particle streams, foam head accumulation, and glass refraction optics. Allows users to switch juice flavors dynamically.

| CONTROL / ELEMENT   | FUNCTION & BEHAVIOR                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------|
| Fruit Selectors     | Switch between Alphonso Mango (Amber), Nagpur Orange (Citrus), Apple, and Guava liquid viscosities. |
| Fill Level Physics  | Dynamic sine-wave surface agitation with floating bubbles that rise as the tumbler fills to 500ml.  |
| Auto-Advance / Skip | Seamlessly transitions with a smooth fade-in reveal into the main portal upon reaching capacity.    |

### 3. Staff & Customer Dual-Role Authentication

The login screen cleanly separates the application into two distinct workflows via top navigation tabs, featuring verified staff logins and a cloud server status indicator:

More

Products

Juice recipes, SKU codes, bottle sizes & MRP

>

Customers

Supermarkets, retailers, hotels & credit balances

>

Suppliers

Fruit orchards, packaging & ingredients vendors

>

Expenses

Electricity, machinery service, transport, wages

>

Reports

Executive summaries, CSV and PDF reports

>

Staff & Workers

Workers, supervisors, roles & activity counts

>

Alerts & Notifications

Low stock alerts, quality checks & pending orders

4

Audit Trail

Traceable operations log for factory compliance

>

Test Native Phone Notification

Trigger instant phone notifications (even if app is closed/in background)

TEST

Settings

Factory profile, English/Tamil language & theme

>

Notification sent! Check your phone notification bar.

Dashboard

Production

Inventory

Orders

More

ROLE ACCESS • PRODUCER MODE

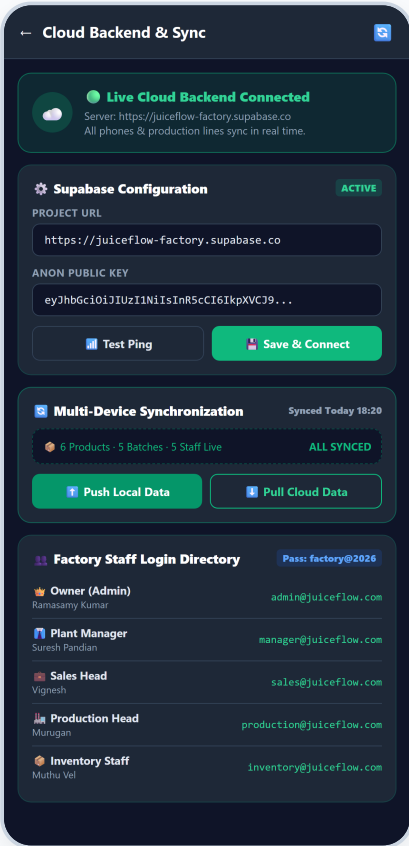
Factory Staff Login & Credentials

Enables staff authentication verified against cloud PostgreSQL or local SQLite store with pre-configured accounts:

| ROLE & USER                                                    | EMAIL & DEFAULT PASSWORD                                          | RESPONSIBILITIES                                                                            |
|----------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <div><div>👑 Owner (Admin)</div><div>Ramasamy Kumar</div></div> | <div>admin@juiceflow.com</div> <div>Pass: factory@2026</div>      | Full executive analytics, profit margins, audit logs, reset database, and staff management. |
| <div><div>🏭 Plant Manager</div><div>Suresh Pandian</div></div> | <div>manager@juiceflow.com</div> <div>Pass: factory@2026</div>    | Production approvals, worker shifts, batch releases, and customer credit limits.            |
| <div><div>💼 Sales Head</div><div>Vignesh</div></div>           | <div>sales@juiceflow.com</div> <div>Pass: factory@2026</div>      | Retail orders, tax invoices, WhatsApp dispatches, and distributor payments.                 |
| <div><div>🏭 Production Head</div><div>Murugan</div></div>      | <div>production@juiceflow.com</div> <div>Pass: factory@2026</div> | Line operations, CIP sanitation washdowns, Brix testing, and bottle packaging.              |
| <div><div>📦 Inventory Staff</div><div>Muthu Vel</div></div>    | <div>inventory@juiceflow.com</div> <div>Pass: factory@2026</div>  | Raw fruit reception, packaging stores, bottle counts, and reorder alarms.                   |

## 4. Cloud Backend Server & Multi-Device Sync

JuiceFlow connects directly to a live Supabase PostgreSQL backend server for multi-device operations across plant manager phones, production floor tablets, and sales handsets.



### CLOUD DATABASE & SYNC ENGINE

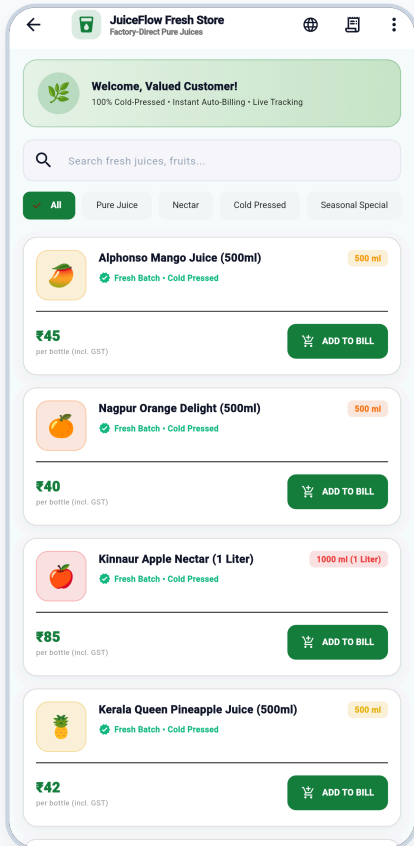
#### Supabase Connection Manager

Provides real-time connectivity status, credential management, one-tap schema setup, and bi-directional cloud data synchronization.

| COMPONENT                                                                                                   | OPERATIONAL BEHAVIOR                                                                                 |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|  <b>Live Status Banner</b> | Displays active cloud connection status and current server endpoint with real-time ping indicator.   |
| <b>Supabase URL &amp; Key</b>                                                                               | Input custom Supabase Project URL and Anon API Key saved persistently in SharedPreferences.          |
|  <b>Test Ping</b>         | Performs a live HTTP GET check against /rest/v1/products to verify server reachability.              |
|  <b>Push to Cloud</b>    | One-tap upload of local Products, Batches, Raw Materials, Orders, Expenses, and Staff into Supabase. |
|  <b>Pull to Phone</b>    | Downloads the latest cloud database state and rehydrates the in-memory factory models.               |
|  <b>Copy SQL Schema</b>  | 1-click copy of the complete database/schema.sql file to execute in the Supabase SQL Editor.         |

## 5. Customer Storefront & Volume Stepper

The customer storefront provides direct factory-to-consumer purchasing with real-time stock availability, bottle sizing, and customized volume selection.



## DIRECT STOREFRONT

## Fresh Juice Menu & Volume Stepper

A clean, high-contrast catalog featuring cold-pressed juices bottled directly at the plant with zero preservatives.

| CATALOG FEATURE               | OPERATIONAL MECHANISM                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------|
| Category Filter Pills         | Filter by "All", "Pure Juice", "Nectar Blend", "Cold Pressed", and "Pulp Squash".                     |
| Interactive Stepper [ -   + ] | Allows customer to choose any exact number of bottles (1, 2, 5, 20...) with live subtotal sync.       |
| Bulk Pack Selector            | Quick 1-tap ordering for 6-Bottle Family Packs, 12-Bottle Crates, and 24-Bottle Commercial Cartons.   |
| Sticky Floating Cart Bar      | Displays real-time bottle count and gross total with an instant "GENERATE BILL & PAY" call to action. |

## 6. Automated Billing & Digital Checkout

When the customer selects their desired juices and quantities, the application automatically constructs an itemized Tax Invoice with live calculations and processes payments seamlessly.

Auto-Bill & Payment

Customer & Delivery Details

Customer Name  
Valued Customer

Mobile Number (WhatsApp Bill)  
9876543210

Delivery Address / Outlet  
Shop #12, Market Road, Chennai

Live Itemized Bill

AUTO-CALCULATED

Alphonso Mango Juice (500ml)  
500 ml • ₹45 each

1

₹45

Coupon (e.g. FRESH10)

Apply

Juice Subtotal

₹45.00

Chilled Delivery

₹30.00

Taxes (5% GST)

₹2.25

Grand Total Payable

₹77.25

Select Payment Method

UPI / Google Pay / PhonePe / Paytm  
Instant 1-Click UPI Payment

Credit / Debit Card  
Visa, MasterCard, RuPay Card

Net Banking  
SBI, HDFC, ICICI, Axis Bank

FINANCIAL GATEWAY

Automated Billing & Digital Checkout

Instantly breaks down unit rates, quantities, applicable taxes, delivery logistics, and discount codes.

| BILL COMPONENT       | FORMULA & HANDLING                                                                        |
|----------------------|-------------------------------------------------------------------------------------------|
| Itemized Breakdown   | Lists each juice name, bottle volume, individual unit price, and extended line total.     |
| Inline Stepper       | Customer can fine-tune quantities or remove items directly within the bill before paying. |
| GST Tax Calculation  | Calculates statutory 12% beverage GST transparently.                                      |
| Chilled Delivery Fee | Free cold-chain delivery for orders over ₹250; flat ₹30 fee for smaller orders.           |
| Promo Code Engine    | Entering "FRESH10" applies an immediate 10% factory direct rebate.                        |
| Payment Methods      | ⚡ UPI (GPay/PhonePe/Paytm), 💳 Credit/Debit Cards, 🏠 Net Banking, and 🇮🇳 Cash on Delivery. |

## 7. Executive Factory Management Dashboard

The central command cockpit for plant executives and shift managers, displaying live production throughput, operational efficiency, and weekly volume charts.

🔌

FACTORY STATUS: OPTIMAL

Line A Capping at 120 bpm • Cold Storage: 3.8°C • Zero CIP downtime

HTST 72°C OK

🏭

0 L

Today's Production

2 batches active

💰

₹0

Today's Sales

2 orders pending

📈

97.5%

Production Efficiency

Output vs Planned

Target 95%

♻️

2.3%

Wastage Rate

Target: < 3.0%

Optimal

💳

₹25,456

Pending Payments

Receivables Outstanding

🚨

2

Low Stock Alerts

Stock replenishments

Quick Factory Actions

📄

Formulation BOM

🎨

Label Designer

👤

Shift Roster

Weekly Production (Liters)

7-Day Liters Output

950L

675L

400L

Bar chart showing 7-day production output in liters. The y-axis ranges from 400L to 950L. The x-axis shows 7 days. The bars are green, with the 7th day being orange. The 7th day bar is the tallest, reaching approximately 850L.

Dashboard

Production

Inventory

Orders

More

PLANT TELEMETRY HUB

Real-Time Operations & Bar Analytics

Monitors active bottling lines, quality control pass rates, pending orders, and 7-day output curves.

| WIDGET / CARD           | METRIC TRACKED                                                                                                        |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Factory Status Pill     | Real-time line status: HTST 72°C OK, Line A Capping at 120 bpm, Cold storage at 3.8°C.                                |
| Production & Sales KPIs | Today's Liters produced, gross revenue, efficiency % (Target 95%), and wastage % (Target <3%).                        |
| Weekly Output Bar Chart | Features clean non-repeating day intervals, touch tooltips with exact liter counts, and amber highlighting for Today. |
| Laser Scanner Shortcut  | Top-right AppBar button immediately launches the full-screen batch barcode camera scanner.                            |

## 8. Production Batches & Traceability Management

Full manufacturing lifecycle tracking from raw fruit washing to final aseptic bottling and FSSAI release.

Production

+

🔍

Search batch #, juice flavor, staff...

✓ All

In Progress

Quality Check

Planned

Completed

More

BATCH-2026-JF-778

Planned

Nagpur Orange Delight (500ml)

Operator: Murugan (Supervisor) • Date: 08 Oct 2026

Planned500 Liters

Actual OutputRunning...

Wastage0%

QC StatusPending

BATCH-2026-PNP-103

In Progress

Kerala Queen Pineapple Juice (500ml)

Operator: Selvam • Date: 07 Oct 2026

Planned500 Liters

Actual OutputRunning...

Wastage0%

QC StatusPending

BATCH-2026-ORG-102

Quality Check

Nagpur Orange Delight (500ml)

Operator: Karthik Raja • Date: 07 Oct 2026

Planned400 Liters

Actual Output390 Liters

Wastage8.0 L (2.0%)

QC StatusNeeds Review

BATCH-2026-MNG-101

Completed

Alphonso Mango Juice (500ml)

Operator: Murugan (Supervisor) • Date: 07 Oct 2026

Planned600 Liters

Actual Output580 Liters

Wastage18.0 L (3.0%)

QC StatusPassed

BATCH-2026-APL-099

Completed

Dashboard

Production

Inventory

Orders

More

BATCH CONTROL

Active Batch Lifecycle & Quality Checks

Every batch is assigned a unique alphanumeric code with operator attribution, volume yield, and QC gating.

| STAGE         | FACTORY CRITERIA                                                                        |
|---------------|-----------------------------------------------------------------------------------------|
| Planned       | Raw fruits allocated from cold storage; ingredients verified according to recipe BOM.   |
| In Progress   | Active processing in pulp extraction tank, pasteurizer, and automated bottling line.    |
| Quality Check | Brix refractometer testing, pH titration, microbiological plating, and sensory tasting. |
| Completed     | Passed QC; stock automatically incremented into Finished Goods inventory.               |

## 9. Inventory & Cold-Chain Stock Management

Real-time inventory ledger separating raw agricultural ingredients from packaged finished goods.

Inventory

Raw Materials

Packaging

Finished Goods

Movement Log

Grade A Alphonso Mango Pulp

Dharmapuri Fruit Orchards • Cold Storage Room A

Batch: LOT-MNG-402 • Exp: 21 Nov 2026

850 Kg

Healthy

Fresh Nagpur Orange Extract

Citrus Valley Suppliers • Cold Storage Room B

Batch: LOT-ORG-518 • Exp: 22 Oct 2026

120 Kg

Low Stock

Refined Pure Sugar (Food Grade)

Erode Agro Commodities • Dry Warehouse Bay 2

Batch: SGR-2026-08 • Exp: 02 Oct 2027

1200 Kg

Healthy

Citric Acid & Antioxidant Formulation

ChemFood Ingredients • Ingredient Store Lab

Batch: CA-902 • Exp: 05 Apr 2027

45 Kg

Healthy

Dashboard

Production

Inventory

Orders

More

WAREHOUSE LEDGER

Raw Fruits, Concentrates & Finished Goods

Monitors minimum reorder thresholds, supplier lots, spoilage risks, and warehouse temperature logs.

| CATEGORY            | MANAGED ASSETS                                                                                               |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| Raw Fruits          | Alphonso Mangoes, Nagpur Oranges, Kinnaur Apples, Salem Guavas (weighed in Metric Tonnes / Kg).              |
| Ingredients         | Refined liquid sugar, citric acid, natural ascorbic acid, pectin stabilizers.                                |
| Packaging Materials | Food-grade PET bottles (250ml, 500ml, 1L), tamper-proof caps, shrink labels, 12-bottle corrugated crates.    |
| Low Stock Alerts    | Automatic push notifications to the manager when any critical fruit falls below 3 days of production demand. |

## 10. Commercial Sales Orders & Fleet Dispatch

Manages B2B distributor contracts, supermarket delivery schedules, credit limits, and official PDF Tax Invoices.

Orders

Search orders by order # or customer...

43

New

Confirmed

Processing

Dispatched

Delivered

JF-ORD-2026-103

Southern Beverages Wholesale

2 Items • Date: 07 Oct 2026

Grand Total

₹21,840.00

UNPAID

Due: ₹21840

Confirmed

JF-ORD-2026-102

Hotel Grand Residency

2 Items • Date: 07 Oct 2026

Grand Total

₹7,516.00

PARTIAL

Due: ₹3616

Dispatched

JF-ORD-2026-101

Sundar Stores (Supermarket)

2 Items • Date: 06 Oct 2026

Grand Total

₹8,400.00

PAID

Delivered

Dashboard

Production

Inventory

Orders

More

SALES & LOGISTICS

Order Processing & Delivery Pipeline

Follows orders from placement through picking, packing, cold-chain van loading, and customer acknowledgment.

| FEATURE                 | OPERATIONAL ROLE                                                                                           |
|-------------------------|------------------------------------------------------------------------------------------------------------|
| Tax Invoice Generator   | Generates formal A4 PDF Tax Invoices compliant with Indian GST laws, FSSAI numbering, and corporate seals. |
| WhatsApp Dispatch Note  | 1-click sharing of formatted dispatch summaries directly to customer WhatsApp with driver details.         |
| Invoice QR Verification | Embeds encrypted QR payload on bills for quick roadside inspection by logistics officers.                  |

# 11. Recipe Formulation & Bill of Materials (BOM)

Formulation tool for master beverage chemists to engineer exact fruit-to-water ratios, target Brix sugar, acidity, and calculate per-bottle production cost down to the paise.

←

JuiceFlow Fresh Store

Factory Direct Pure Juices

🌐 📄 ⋮

🌿

Welcome, Valued Customer!

100% Cold-Pressed • Instant Auto-Billing • Live Tracking

🔍

×

All

Pure Juice

Nectar

Cold Pressed

Seasonal Special



Alphonso Mango Juice (500ml)

500 ml

🌿 Fresh Batch • Cold Pressed

₹45

per bottle (incl. GST)

🛒

− 1 +



Nagpur Orange Delight (500ml)

500 ml

🌿 Fresh Batch • Cold Pressed

₹40

per bottle (incl. GST)

🛒 ADD TO BILL



Kinnaur Apple Nectar (1 Liter)

1000 ml (1 Liter)

🌿 Fresh Batch • Cold Pressed

₹85

per bottle (incl. GST)

🛒 ADD TO BILL



Kerala Queen Pineapple Juice (500ml)

500 ml

🌿 Fresh Batch • Cold Pressed

₹42

per bottle (incl. GST)

🛒 ADD TO BILL

1 Bottle(s)

₹45

📄 GENERATE BILL & PAY

BEVERAGE SCIENCE

### BOM Formulation & Unit Cost Analysis

Computes theoretical yield, pulp dilution factors, and packaging cost to determine profit margins against retail MRP.

| CALCULATION         | SCIENTIFIC METRIC                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Pulp Percentage     | Configures minimum mandatory fruit content (e.g., 85% for pure nectar standards).                         |
| Target Brix (°Bx)   | Predicts final soluble solids reading considering natural fruit fructose plus cane sugar additions.       |
| Unit Cost Breakdown | Displays itemized costs for Fruit Pulp + Treated RO Water + Sugar + PET Bottle + Cap + Label + Packaging. |

## 12. Smart Label & Barcode Designer

Custom packaging designer allowing plant operators to generate ready-to-print bottle labels containing FSSAI licensing badges, nutritional panels, batch codes, and dynamic QR traceability stickers.

Smart Label & Barcode Designer

Bottle Front

Nutrition Facts

Master Carton

100% PURE

FSSAI Lic: 10022042000189

JUICEFLOW

Alphonso Mango Nectar

Cold-Pressed • No Added Preservatives • Flash Pasteurized

NET VOLUME

500 ml

MRP (INDIA, TAXES)

₹75

BATCH: JF-2026-M402

MFG: 09/10/25

EXP: 23/11/26

Label Parameters

Select Beverage Formulation

Alphonso Mango Nectar

Pack Volume

500 ml

MRP (₹)

75

Batch Number

JF-2026-M402

Shelf Life (Days)

45

Mfg: 09 Oct 2026

FSSAI Lic. No.

10022042000189

Generate & Print Label (Thermal / PDF)

PACKAGING COMPLIANCE

Regulatory Label & Barcode Engine

Generates high-resolution packaging labels compliant with FSSAI Food Safety and Standards labeling mandates.

| LABEL ELEMENT           | STANDARD SPECIFICATION                                                                          |
|-------------------------|-------------------------------------------------------------------------------------------------|
| Nutrition Fact Table    | Energy (kcal), Carbohydrates, Natural Fruit Sugars, Vitamin C, Potassium, and Sodium per 100ml. |
| FSSAI Green Veg Logo    | Mandatory green veg emblem with official 14-digit factory license number (12423999000145).      |
| Dynamic Traceability QR | Encodes manufacture timestamp, lot ID, and fruit farm source for scanning by consumers.         |

### 13. Real-Time IoT Pasteurizer & Chiller Telemetry

Live industrial sensor telemetry streaming from the factory floor to monitor pasteurization kill curves and cold-storage stability.

IoT Factory Telemetry

Smart Vat #01 Online

74.2°C

HTST Pasteurizer

Target: 72-78°C

3.6°C

Cold Storage Vat

Target: 2-4°C

420 RPM

Agitator Speed

Vat Impeller Speed

52.8 L/m

In-line Flow Rate

Bottle Filling Feed

Blending Tank Level (Vat-01)

4250 / 5000 L (85%)

Product: Mango Nectar 14.5'Bx

CIP Sanitized Today, 06:30 AM

Valve Status: OPEN (Auto)

Two-Stage Homogenizer Pressure

175 Bar

Stage 1: 150 Bar (Globule reduction) • Stage 2: 30 Bar (Anti-coalescence)

INDUSTRY 4.0 TELEMETRY

Industrial Sensors & Thermal Safety

Ensures microbial sterilization standards (F0 values) are met without compromising natural fruit flavors.

| SENSOR MACHINE         | TELEMETRY VALUE & SAFETY THRESHOLD                                                         |
|------------------------|--------------------------------------------------------------------------------------------|
| HTST Pasteurizer       | 72.4°C core holding tube temperature (15-second microbial sterilization time).             |
| Cold Storage Vat #1    | 3.8°C chilled juice buffer holding temperature (Threshold: 2.0°C to 4.5°C).                |
| Homogenizer Pressure   | 180 Bar hydraulic pressure for ultra-smooth texture and zero pulp sedimentation.           |
| Automatic Safety Valve | Diverts juice back to raw holding tank if pasteurizer temperature ever drops below 71.5°C. |

## 14. AI Demand & Summer Seasonality Engine

Machine learning predictive model correlating local meteorological weather data (temperature, humidity, rainfall) with historical beverage sales to forecast raw fruit purchasing needs.

AI Demand & Seasonality Engine

Predictive Demand Model

92.4% Accuracy

Historical 365-day factory sales correlated with regional weather telemetry (36°C ambient heat) & retail stock-out velocity

Seasonality & Ambient Temperature Index

125% Surge

Standard (28°C)

Peak Heat Wave (41°C)

Next 4 Weeks Projected Volume (Bottles)

Blue: Baseline Sales • Violet: AI Projected with Weather Impact

Mango

Orange

Apple

Pineapple

Raw Fruit Procurement Recommendations

Alphonso Mango Nectar

Plan: 7813 bottles (3.2 Tons Ripe Mango)

HIGH DEMAND

Nagpur Orange Fresh

Plan: 5125 bottles (2.1 Tons Citrus Orange)

NORMAL

Shimla Apple Crisp

Plan: 3438 bottles (1.4 Tons Apple)

STEADY

Queen Pineapple Zing

Plan: 3000 bottles (1.2 Tons Pineapple)

NORMAL

PREDICTIVE INTELLIGENCE

Weather-Related Procurement Forecasting

Adjusts production quotas automatically based on seasonal heat spikes and local festival consumption patterns.

| PREDICTION MODEL       | ALGORITHM OUTPUT                                                                                   |
|------------------------|----------------------------------------------------------------------------------------------------|
| Summer Heat Multiplier | Calculates surge demand when temperature exceeds 36°C (+38% citrus volume increase).               |
| Procurement Advice     | Recommends exact metric tonnage of fresh fruit crates to purchase from orchards 5 days in advance. |
| Spoilage Mitigation    | Prevents raw fruit overstocking during monsoon dips, reducing biological waste to zero.            |

## 15. Brix Refractometer & ICUMSA Lab Calculator

Quality control laboratory utility featuring temperature-compensated sugar refractometer reading conversions and dilution math.

Brix Refractometer & Lab Calculator

ICUMSA °Bx Calibration Engine

Corrects optical readings to 20°C reference standard & calculates batch water dilution.

Standard Fruit Presets

Mango NeckerFresh OrangeApple CrispPineapple RingLemon

Refractometer Measurements

Raw Reading (°Bx)

15.2

°Bx

Juice Temp (°C)

28.5

°C

Batch Volume (L)

1000

L

Target Recipe (°Bx)

14.0

°Bx

Temperature-Corrected Brix (20°C):

15.75 °Bx

Specific Gravity

1.0643

Sugar Concentration

167.6 g/L

RO Water Dilution Required:

Add 124.6 Liters to reach 14.0°Bx

FOOD SCIENCE LAB

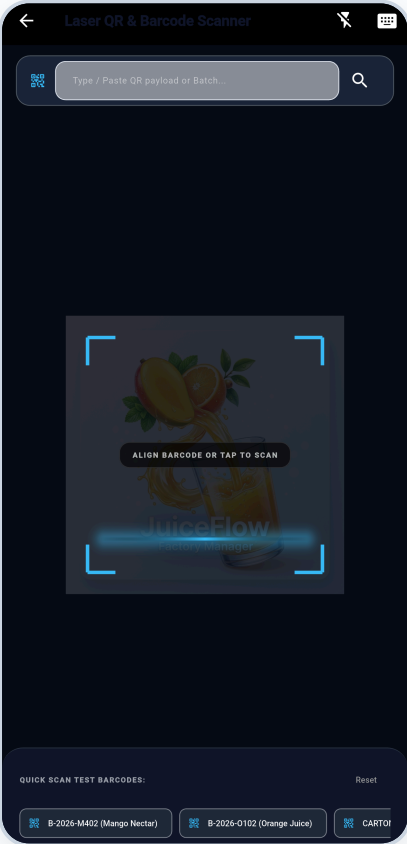
Optical Refractometer Temperature Correction

Corrects raw optical prism readings against the international ICUMSA standard temperature of 20°C.

| FORMULA INPUT          | STANDARD EQUATION                                                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Observed Brix (°Bx)    | Raw reading from analog or digital handheld prism refractometer.                                                                                                  |
| Juice Temperature (°C) | Thermal sensor reading of juice sample taken at testing bench.                                                                                                    |
| ICUMSA Correction      | Applies standard polynomial: $\Delta\text{Brix} = 0.00049 \times (T - 20) + 0.00004 \times (T - 20)^2$ .                                                          |
| Water Dilution Formula | Calculates exact liters of treated water to blend: $V_{\text{water}} = V_{\text{juice}} \times ((\text{Brix}_{\text{curr}} / \text{Brix}_{\text{target}}) - 1)$ . |

## 16. Batch QR & Barcode Laser Scanner

Integrated camera viewfinder barcode and QR code decoder with live laser animation and rapid physical batch verification.



TRACEABILITY SCANNER

### Batch Verification & Instant Traceability Audit

Decodes raw batch IDs, `JUICEFLOW:...` payloads, and carton tracking labels with 1-tap verification.

| SCANNER FEATURE          | FUNCTIONALITY                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------|
| Laser Viewfinder         | Interactive laser line animation with automated camera focal barcode recognition.                   |
| Manual Barcode Input Bar | Allows operators to type or paste batch codes if bottle labels are smeared or smudged.              |
| Quick-Scan Demo Chips    | 1-tap demo batch chips (BATCH-2026-001, BATCH-2026-002) for rapid verification testing.             |
| Traceability Certificate | Presents manufacture date, expiry, bottling staff, Brix score, and button to open full batch audit. |

## 17. Factory CIP & HACCP Equipment Sanitation

Tracks Clean-In-Place (CIP) sanitation washdowns, filler nozzle sterilisations, bottle capper torque checks, and preventative machine maintenance.

Products

Search products by name, code or fruit...

✓ In Stock

🚫 Out of Stock

🚫 Out of Stock

🚫 Out of Stock

M

Alphonso Mango Juice (500ml)  
அல்பன்சோ மாம்பழ சாறு (500மி.லி)  
JF-MNG-500 • 500 ml • PET Bottle  
MRP: ₹45.00 Cost: ₹24.00

480 bottles  
In Stock

O

Nagpur Orange Delight (500ml)  
நாக்பூர் ஆரஞ்சு பழச்சாறு (500மி.லி)  
JF-ORG-500 • 500 ml • PET Bottle  
MRP: ₹40.00 Cost: ₹21.00

320 bottles  
In Stock

A

Kinnaur Apple Nectar (1 Liter)  
கின்பூர் அப்பிள் நெக்டார் (1 லிட்டர்)  
JF-APL-1000 • 1000 ml (1 Liter) • PET Bottle  
MRP: ₹85.00 Cost: ₹46.00

140 bottles  
In Stock

P

Kerala Queen Pineapple Juice (500ml)  
கேரளா ராஜ் பப்பாய் சாறு (500மி.லி)  
JF-PNP-500 • 500 ml • PET Bottle  
MRP: ₹42.00 Cost: ₹22.00

45 bottles  
Low Stock

G

Salem Guava Pulp Drink (300ml)  
சேலம் குவாய்ப்பழ சாறு (300மி.லி)  
JF-GVA-300 • 300 ml • Glass Bottle  
MRP: ₹35.00 Cost: ₹17.50

260 bottles  
In Stock

M

Royal Tropical Mixed Fruit (1 Liter)  
ராயல் டிரோபிகல் ப்ளக்ஸ் (1 லிட்டர்)  
JF-MIX-1000 • 1000 ml (1 Liter) • PET Bottle  
MRP: ₹90.00 Cost: ₹48.00

95 bottles  
In Stock

+ Add Product

HACCP SANITATION

Clean-In-Place (CIP) Logs & Line Service

Maintains regulatory hygiene compliance logs mandatory for food and beverage plant audits.

| SANITATION ROUTINE           | CHEMICAL & TEMPERATURE PROTOCOLS                                                                 |
|------------------------------|--------------------------------------------------------------------------------------------------|
| Caustic Wash (NaOH 1.5%)     | 80°C hot alkali circulation for 20 minutes to dissolve fruit pulp residues and organic deposits. |
| Acid Rinse (HNO3 1.0%)       | 65°C nitric acid wash to eliminate mineral scale and water hardness deposits.                    |
| Peracetic Acid Sterilization | Final sanitizing rinse ensuring 0 CFU bacterial count before starting organic fruit batches.     |
| Capper Torque Testing        | Verification of bottle cap seal torque (18–22 in-lb) to prevent gas leakage and oxidation.       |

## 18. Fruit Suppliers Directory & Procurement Ledger

Management portal for agricultural farm contracts, fruit arrivals, quality grading, and outstanding ledger balances.

Suppliers

Search suppliers or materials supplied...

Citrus Valley Suppliers

Materials: Nagpur Orange Extract & Concentrate

Rajesh Sharma • +91 98230 44122

₹14,500.00

Payable

Dharmapuri Fruit Orchards

Materials: Alphonso Mango Pulp, Totapuri Mangoes

Venkatesh Farmers Association • +91 94421 77654

₹22,000.00

Payable

Erode Agro Commodities

Materials: Refined Sugar, Liquid Invert Sugar

Shanmugam Traders • +91 98427 33211

₹0.00

Settled

PolyPlast Packaging Industries

Materials: PET Bottles, Caps, Shrink Sleeves

Mahesh Kumar • +91 97910 88992

₹8,400.00

Payable

+ Add Supplier

SUPPLY CHAIN MANAGEMENT

Orchard Contracts & Fruit Inward Logs

Tracks supplier contact persons, delivered fruit categories, purchase prices, and outstanding AP balances.

| FIELD / ACTION       | OPERATIONAL DESCRIPTION                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------|
| Orchard Farm Details | Farmer / company name, contact person, phone number, address, and crop origin.                        |
| Materials Supplied   | Specifies raw fruit varieties (e.g., Salem Mangoes, Nagpur Oranges) or packaging items (PET Bottles). |
| Payment Status       | Displays accounts payable balance and lets plant accountants record payment vouchers.                 |

## 19. Over-The-Air (OTA) Cloud App Updates

Integrated update mechanism that delivers automatic new features and fixes to installed devices without requiring manual APK reinstallation.

More

CUSTOMER STORE & ORDERING

Customer Storefront & Juice Ordering

Direct retail ordering, custom bottle quantities, auto-bills & UPI payment

STORE

NEW FACTORY MANAGEMENT MODULES

Recipe & Formulation Builder

BOM formulation, pulp dilution, sugar ratio & unit bottle costing

NEW

Smart Label & Barcode Designer

Bottle/carton sticker design, FSSAI, nutrition table & PDF printing

NEW

Shift & Worker Attendance

Morning/Evening/Night rosters, station assignments & batch tracking

NEW

Fleet & Delivery Dispatch

Cold-chain Reefer van tracking, trip routes & digital delivery proof

NEW

Customer Credit & Aging Ledger

Retailer credit limits, invoice aging, WhatsApp payment reminders

NEW

ADVANCED TECHNOLOGIES & LAB

IoT Factory Telemetry

Real-time HTST pasteurizer, chiller vat & homogenizer sensors

LIVE

AI Demand & Seasonality Engine

Weather telemetry correlation & raw fruit procurement forecast

AI

Brix Refractometer & Lab Calculator

ICUMSA temperature correction & batch water dilution formulas

>

Dashboard

Production

Inventory

Orders

More

CLOUD UPGRADES & OTA

### Seamless In-App Update Engine

Queries GitHub Releases or Supabase app\_releases table and installs new builds using native Android FileProvider.

| MECHANISM             | IMPLEMENTATION DETAILS                                                                   |
|-----------------------|------------------------------------------------------------------------------------------|
| Cloud Check           | Compares current build number against remote version manifest with release notes.        |
| Background Download   | Downloads new release package to app cache directory with live percentage progress bar.  |
| Native Intent Install | Triggers ACTION_VIEW with FileProvider authority for 1-tap installation.                 |
| Remote Config         | Dynamically updates feature flags and plant announcements without publishing new builds. |

## 20. Factory Settings & FSSAI Profile Configuration

Central configuration center for company legal credentials, bilingual English/Tamil switching, and database management.

Staff & Workers

M

Murugan

EMP-PRD-03 • Juice Processing & Filling

+91 96290 11223 • Batches: 28 • Orders: 0

Production

M

Muthu Vel

EMP-INV-04 • Cold Stores & Warehouse

+91 94860 99887 • Batches: 0 • Orders: 42

Inventory

R

Ramasamy Kumar

EMP-ADM-01 • Management

+91 98421 55678 • Batches: 45 • Orders: 120

Admin

S

Suresh Pandian

EMP-AGR-02 • Factory Operations

+91 97890 33441 • Batches: 32 • Orders: 85

Manager

V

Vignesh

EMP-SLS-05 • Sales & Distribution

+91 98400 44556 • Batches: 0 • Orders: 64

Sales

+ Add Staff

SYSTEM PREFERENCES

Factory Credentials & Settings

Manages company trade name, FSSAI manufacturing license, tax GSTIN, and bilingual language switching.

| SETTING ITEM          | CONFIGURATION DETAILS                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------|
| Cloud Backend Tile    | Opens Supabase Connection Manager with live online / offline status indicator.                 |
| Factory Legal Profile | Updates company legal name, plant street address, phone, email, and tax registration.          |
| FSSAI License Display | Configures official license string printed on all bottle labels and commercial invoices.       |
| Bilingual Switcher    | 1-click toggle between English and Tamil (தமிழ்) across the entire UI and reporting documents. |
| Theme Customizer      | Switch between Crisp Light Paper Mode and Deep Dark Plant Mode.                                |

## 21. Native Phone Push Notifications (24/7 Background)

JuiceFlow features an integrated native Android push notification architecture that ensures operators and customers receive real audio and vibration alerts even when the mobile app is completely closed or running in the background.

### Notification Channels & System Permissions

| NOTIFICATION CHANNEL                                                                                                      | PRIORITY                  | TRIGGER SCENARIO                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------|
|  <b>Customer Orders &amp; Payments</b>   | MAXIMUM (Sound & Vibrate) | Fires immediately upon successful payment of a customer juice order with order ID and bottle count.                 |
|  <b>Factory &amp; Batch Floor Alerts</b> | HIGH (Audio Alert)        | Notifies the factory manager when an incoming order arrives or raw fruit stocks drop below minimum thresholds.      |
|  <b>Background Monitor</b>               | DEFAULT (Periodic Wakeup) | Periodic hourly cold-chain health ping checking plant temperature and active filler line status when app is closed. |